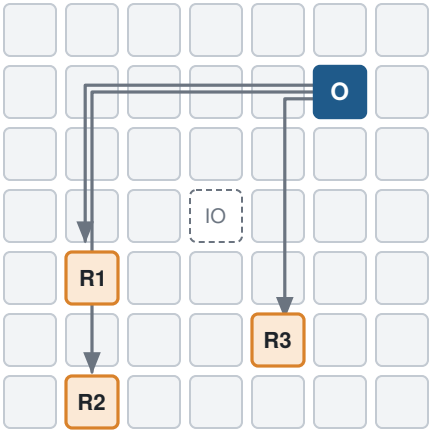
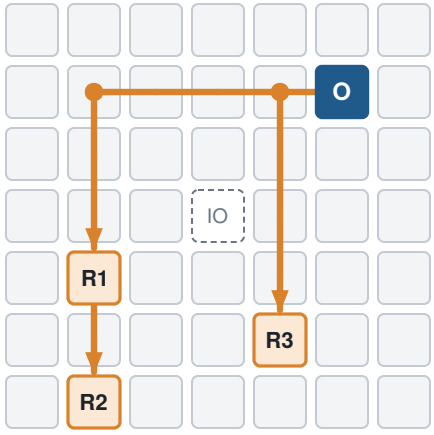
(a) Baseline



owner reads the line 3 times,
sends 3 separate responses

(b) Owner-side cross-GPM merge



owner reads once; one response forks along the routing tree (dots)

(c) Table entries (cuckoo-indexed)

Remote request tracker (RRT), per GPM: 64 → 100s–1000s

fingerprint	dst GPM	waiter list ptr	V

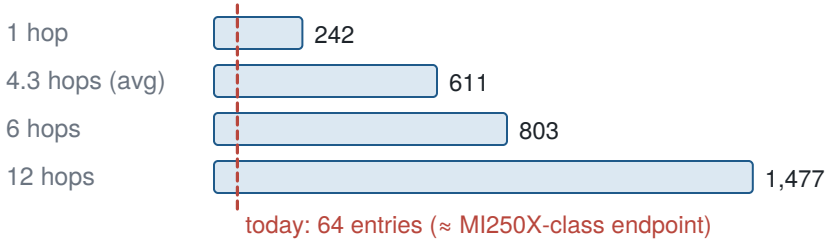
\+ full line tag kept in a side array: checked before any merge

Owner merge table (OMT), per GPM [new]

fingerprint	requester bitmap (48 b)	state

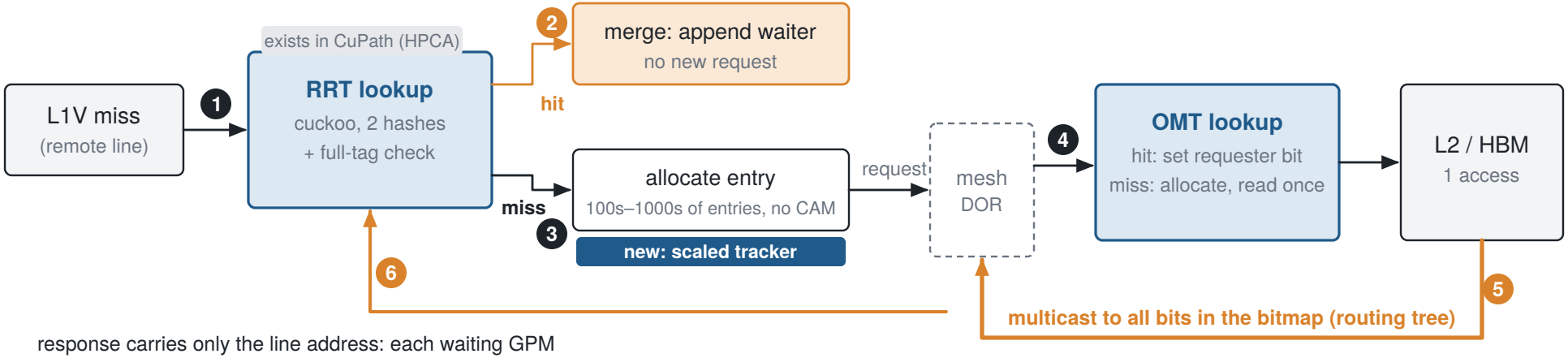
Why the tracker must grow on a wafer

entries needed for 100 B/ns remote = 100 × RTT / 64 B



(d) Requester side (network interface of each GPM)

(e) Owner side [new]



response carries only the line address: each waiting GPM
claims it in its RRT, wakes the waiters, frees the entry